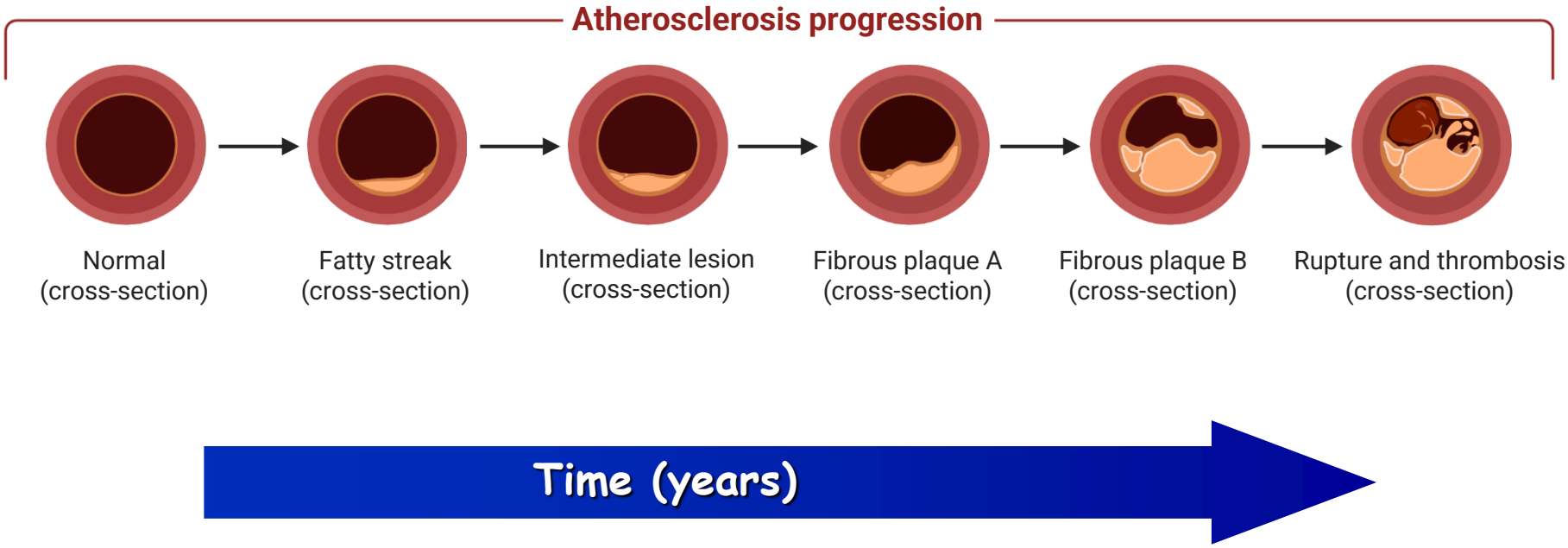
Clínica Universidad de Navarra

Principles of Laboratory Medicine

Dr. Nerea Varo
Clinical Biochemistry Laboratory
Clínica Universidad de Navarra

Acute myocardial infarction

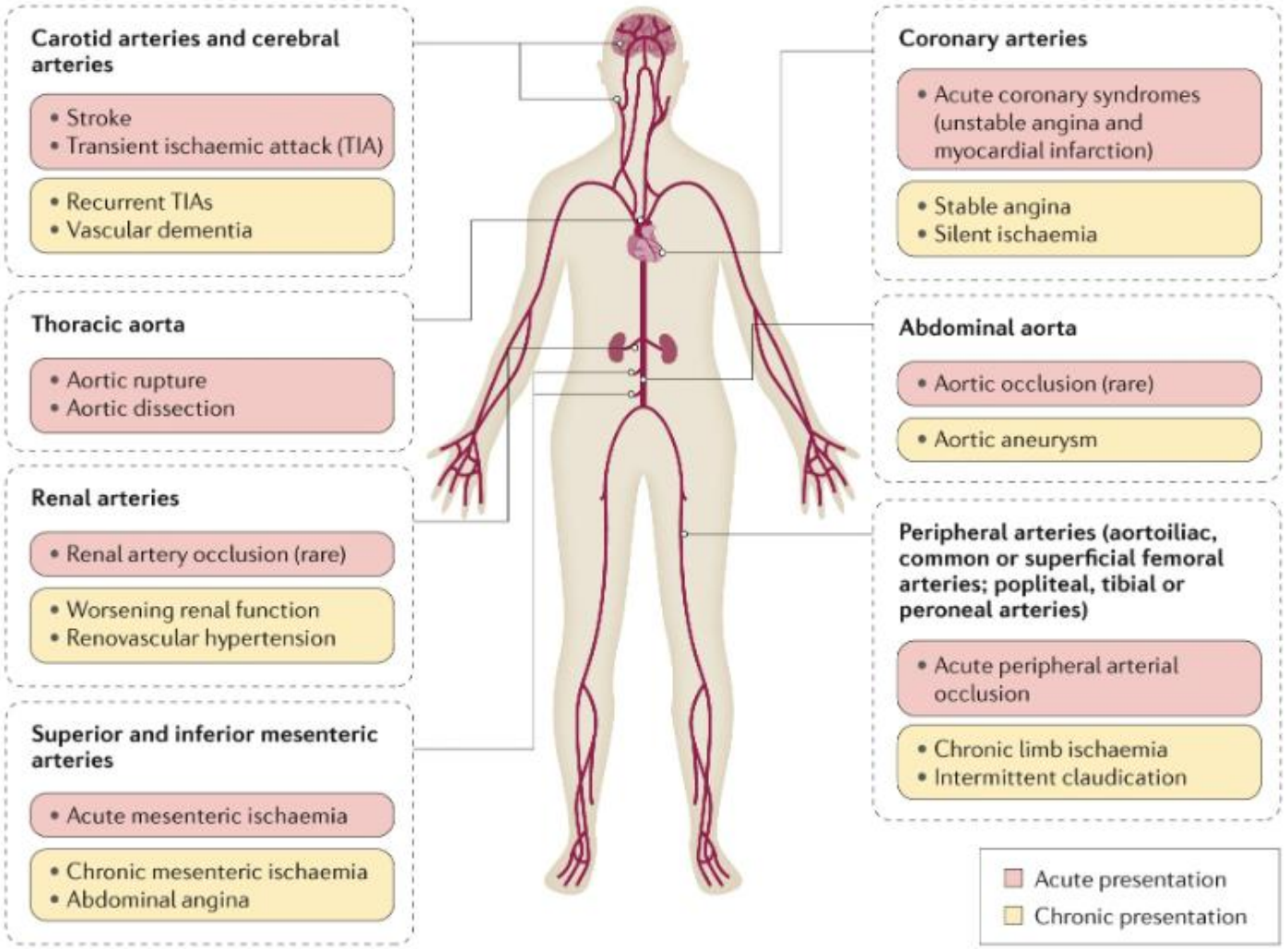
Atherogenesis



Chronic progressive inflammatory process

Fig. 5: Clinical manifestations of atherosclerosis.

From: Atherosclerosis



Atherosclerosis is a systemic disease that may involve multiple vessels. Consequently, the clinical manifestations vary widely according to the vascular territory involved. Despite the systemic nature of many risk factors such as hypercholesterolaemia, hypertension, diabetes mellitus and smoking, atherosclerosis tends to involve primarily specific regions of the arterial tree. Arterial areas subjected to either disturbed flow or low shear stress have particular susceptibility to atheroma formation[41]. These conditions prevail at branch points in the arterial tree.

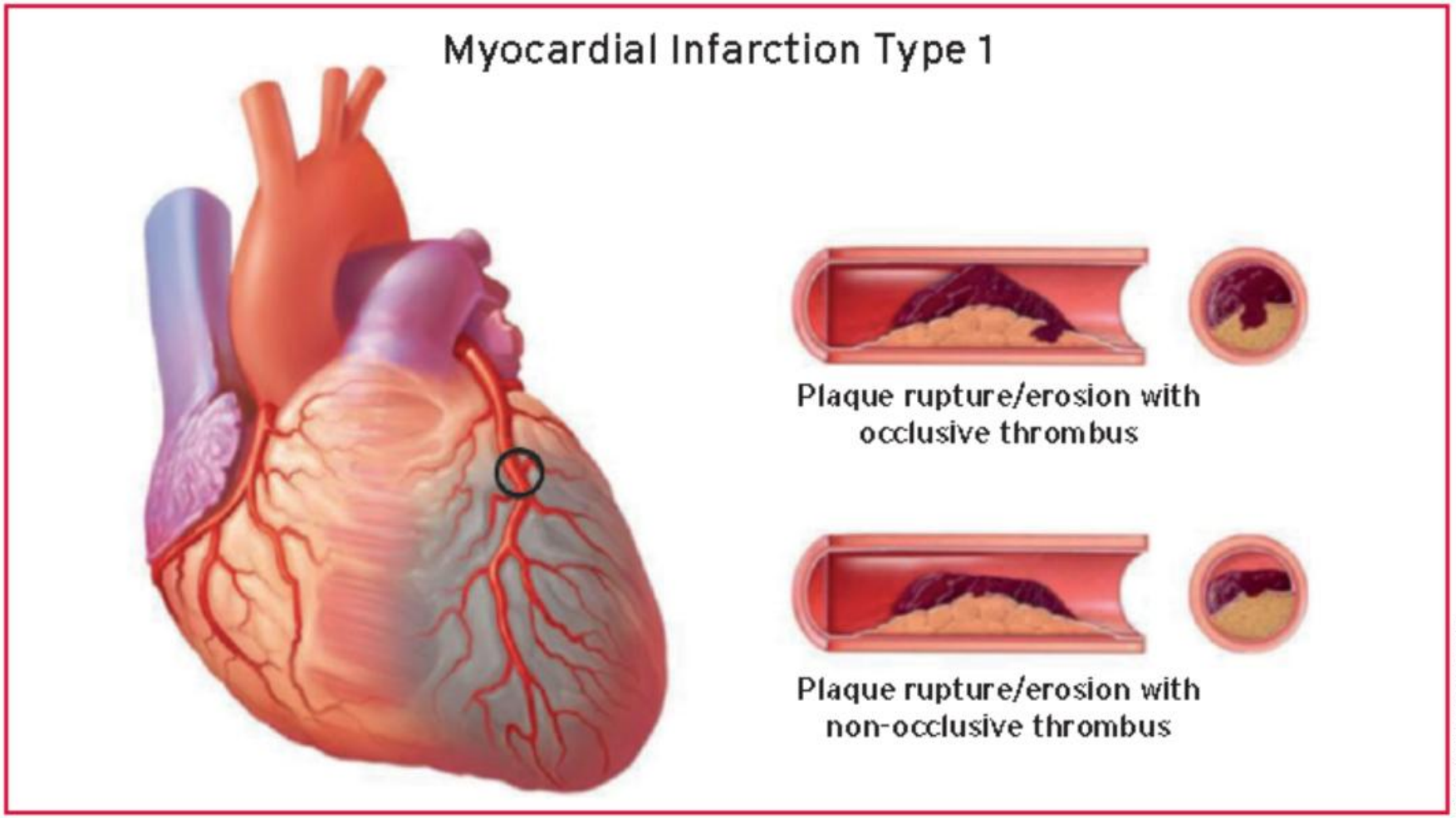
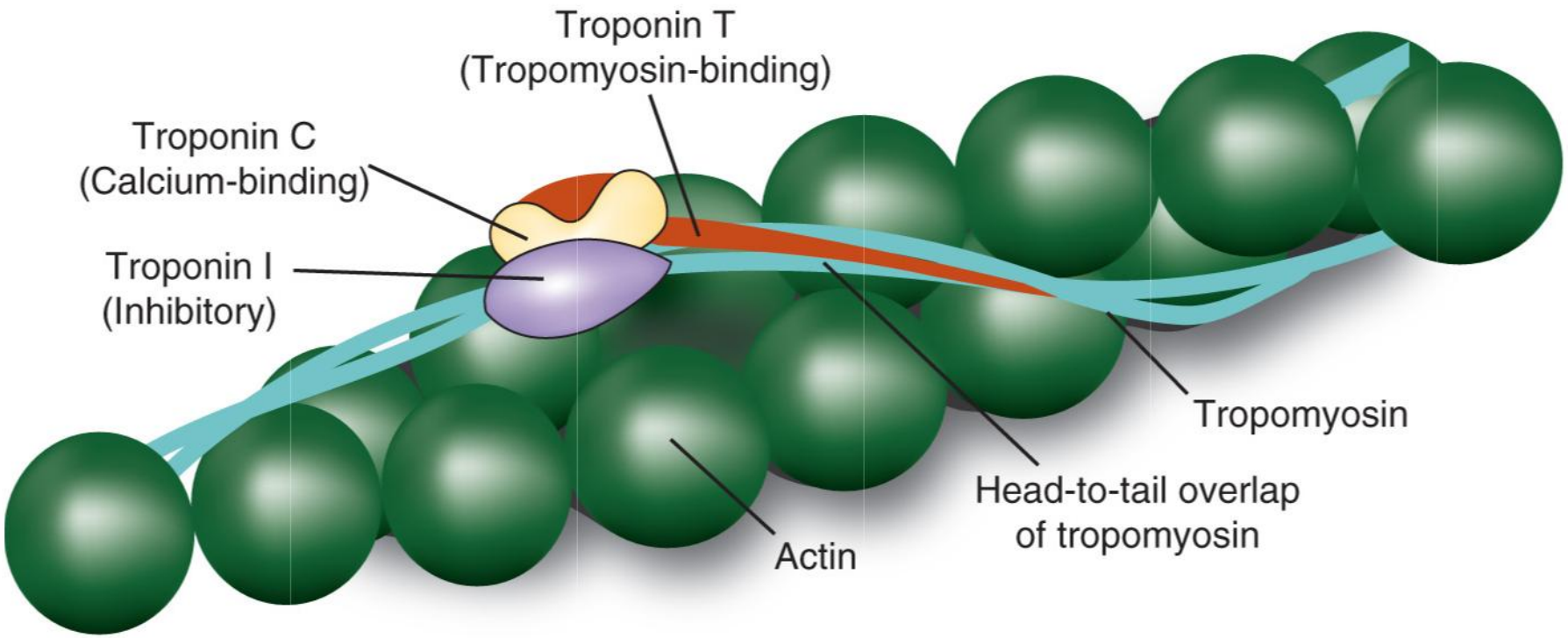


Figure 3 Myocardial infarction type 1.

Sarcomere structure
Troponins



High sensitivity cardiac troponin

UNIVERSAL DEFINITION OF MYOCARDIAL INFARCTION ESC 2018

Universal definitions of myocardial injury and myocardial infarction

Criteria for myocardial injury

The term myocardial injury should be used when there is evidence of elevated cardiac troponin values (cTn) with at least one value above the 99th percentile upper reference limit (URL). The myocardial injury is considered acute if there is a rise and/or fall of cTn values.

Criteria for acute myocardial infarction (types 1, 2 and 3 MI)

The term acute myocardial infarction should be used when there is acute myocardial injury with clinical evidence of acute myocardial ischaemia and with detection of a rise and/or fall of cTn values with at least one value above the 99th percentile URL and at least one of the following:

- Symptoms of myocardial ischaemia;
- New ischaemic ECG changes;
- Development of pathological Q waves;
- Imaging evidence of new loss of viable myocardium or new regional wall motion abnormality in a pattern consistent with an ischaemic aetiology;
- Identification of a coronary thrombus by angiography or autopsy (not for types 2 or 3 MIs).

Post-mortem demonstration of acute athero-thrombosis in the artery supplying the infarcted myocardium meets criteria for *type 1 MI*.
Evidence of an imbalance between myocardial oxygen supply and demand unrelated to acute athero-thrombosis meets criteria for *type 2 MI*.
Cardiac death in patients with symptoms suggestive of myocardial ischaemia and presumed new ischaemic ECG changes before cTn values become available or abnormal meets criteria for *type 3 MI*.

Criteria for coronary procedure-related myocardial infarction (types 4 and 5 MI)

Criteria for myocardial injury

Detection of an elevated cTn value above the 99th percentile URL is defined as myocardial injury. The injury is considered acute if there is a rise and/or fall of cTn values.

Clinical criteria for MI

The clinical definition of MI denotes the presence of acute myocardial injury detected by abnormal cardiac biomarkers in the setting of evidence of acute myocardial ischaemia.

Why the clinical limit of 99th percentile?

- Negative predictive value (rule-out myocardial infarction) >98%
- Positive predictive value (rule-in myocardial infarction) 75-80%

Troponins in myocardial injury

1. Diagnosis
2. Prognosis
3. Monitoring

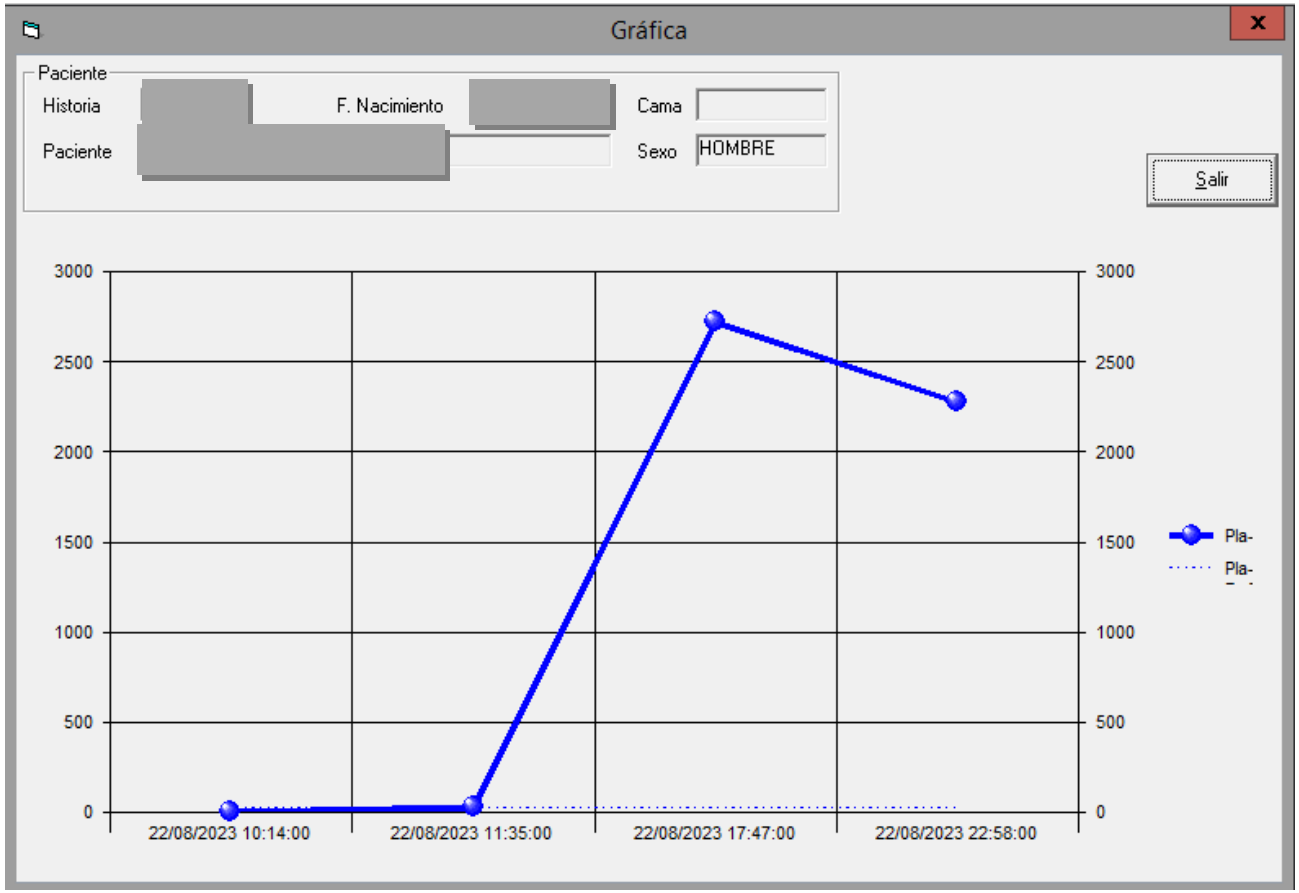
Analytical phase
high sensitivity troponins

1) the total imprecision (CV) at the 99th percentile value should be ≤10%

and

2) measurable concentrations below the 99th percentile should be attainable for at least 50% of healthy individuals

Varón de 56 años que hacia las 09:15 comienza con dolor retroesternal opresivo de intensidad progresivamente creciente, con irradiación y sensación de pesadez en ambos brazos. Acude a urgencias, donde se realiza ECG a las 09:53 en ritmo sinusal con elevación del segmento ST en cara lateral. Se activa código infarto.



Diagnóstico: **cardiopatía isquémica**
Se le realiza cateterismo, observándose un trombo en arteria segunda marginal

Learning goals

1. Introduction to plasma enzymes
2. Measurement of serum enzyme activity
    1. Preanalytical factors
    2. Methods
    3. Interpretation of results
3. Plasma enzymes in disease
    1. Liver: ALT, AST, γ-GT, alkaline phosphatase, LDH
    2. Muscle: creatine kinase, aldolase
    3. Pancreas: amylase and lipase
4. Acute myocardial infarction

1. Introduction to plasma enzymes

Circulating enzymes

- Physiologic function in blood
- Cell remodelling
- Tissue damage

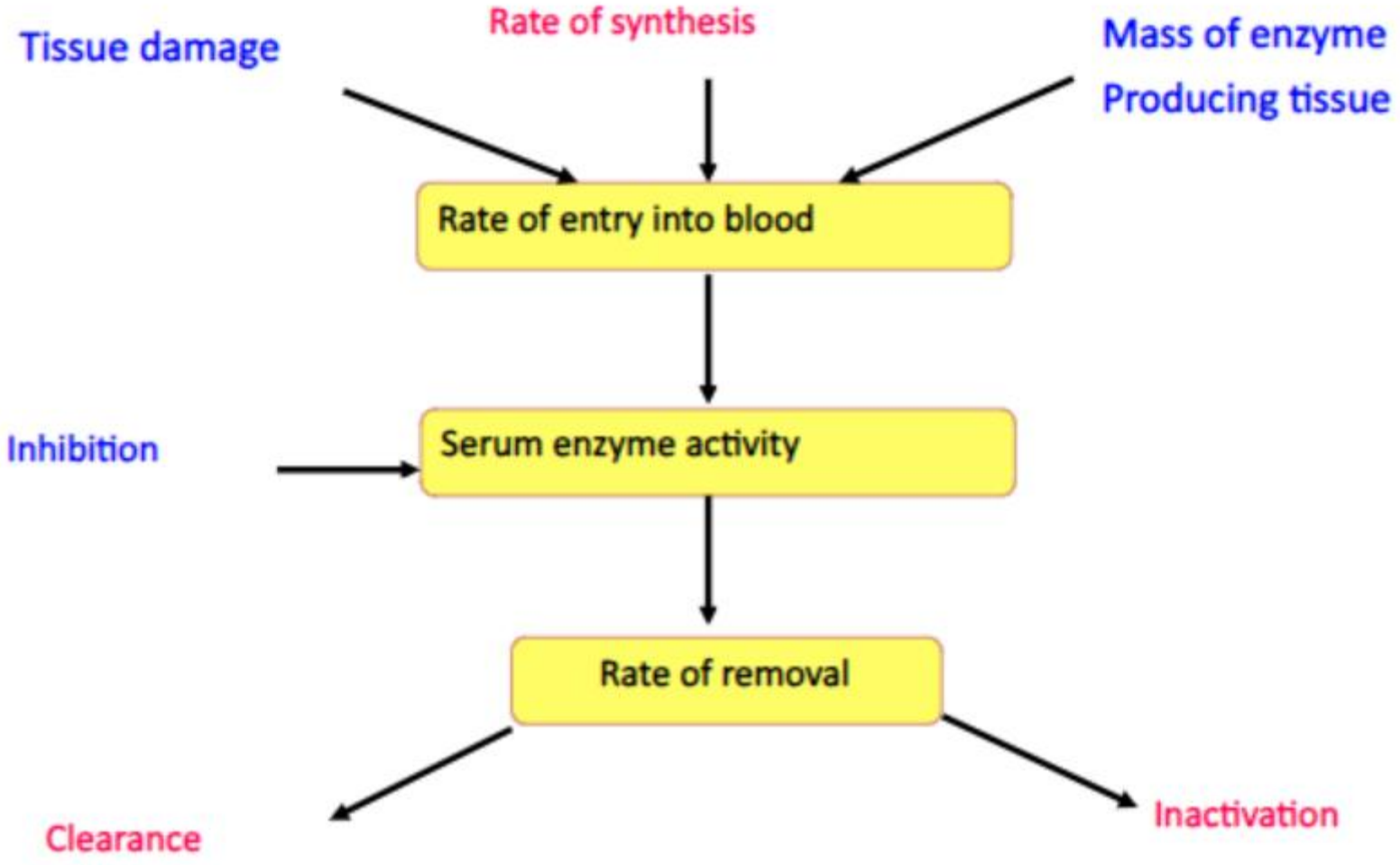
TABLA 17-1 Actividad enzimática relativa respecto al suero

	AST	ALT	CK
Hígado	7.100	2.850	1
Músculo esquelético	5.000	300	50.000
Corazón	7.800	450	10.000
Riñón	4.500	120	640
Eritrocitos	17	7	—

ALT: alanina-aminotransferasa; AST: aspartato-aminotransferasa; CK: creatina-cinasa.

Useful to identify the damaged organ

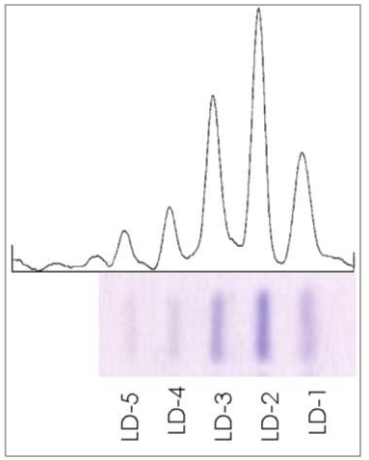
Factors that determine plasma enzymes activity



Isoenzymes

Definition: Different enzymes molecules with the same activity

- Amylase
- Creatine quinase (CK)
- Alkaline phosphatase
- Lactate dehydrogenase (LDH)



Method: electrophoresis

- Time consuming
- Trained personnel
- Expert interpretation

Enzyme quantitation

-Total enzyme:

- quantifies the total amount of the enzyme in blood.

-The test can't tell the difference between the types of isoenzymes.

Does not giveinformation of with tissue may be damaged.

- It is not a specific test.

Isoenzyme test

- measures the amount of each type of isoenzyme in blood.

- helps to figure out which tissues may be damaged.

2. Measurement of serum enzyme activity

2. Measurement of serum enzyme activity

1 Preanalytical factors

2 Analytical Methods

3 Postanalytical: Interpretation of results

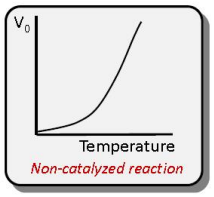
Improve diagnostic performance

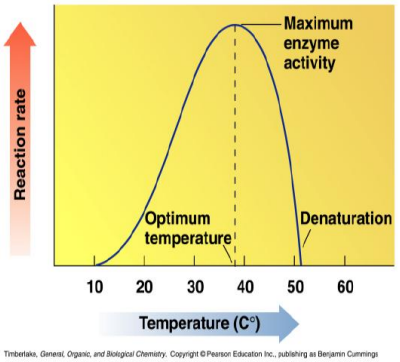
2.1 Determination of Enzyme activity Preanalytical factors

1. Age (Alkaline phosphatase
2. Sex (creatine kinase, transaminase)
3. Physical exercise (creatine kinase
4. Time of sample collection

2.2. Methods

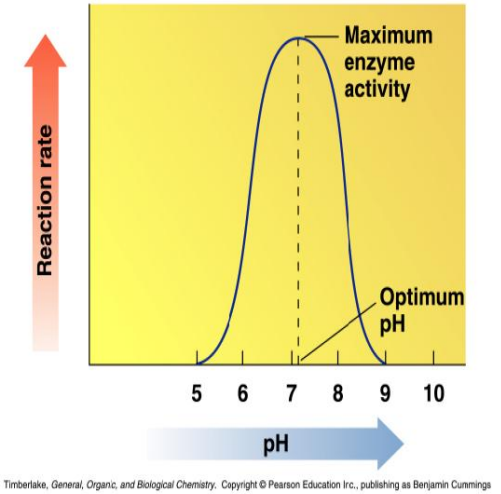
Effect of temperature





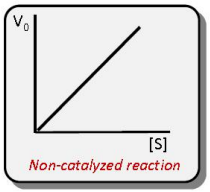
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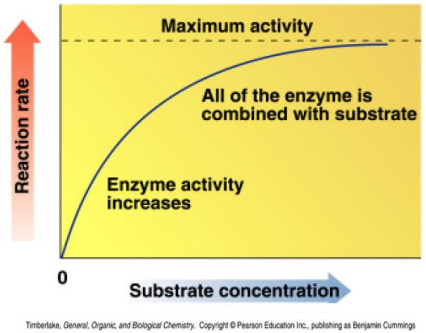
Effect of pH



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Effect of substrate concentration





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Enzymes can be saturated with substrate

Pyridoxal phosphate (PLP)

- Derived from vitamin B_6
- Transaminations, isomerizations, modifications of amino acid side chains, decarboxylations, etc.

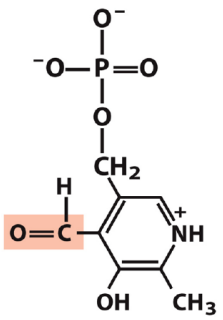


Figure 18-5a
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2. 2. Methods

- Enzyme reaction depends on:
  - Enzyme concentration
  - Substrate type and concentration
  - Incubation temperature (30º, 37º)
  - pH
  - cofactors or inhibidors
- Measurement units
  - Catalytic activity
    - katal: mols of substrate transformed per second
    - IU: µmols of substrate transformed per min **(µmoles/minuto**).
  - Enzyme concentration
    - katal/L
    - UI/L. (Conversión 1 µkat/L = 60 UI/L)

Laboratory report

- Reference range
  - Age
  - Gender

- Temperatura

- Units

Interpretation of results
2.3 Improve diagnostic performance of serum enzyme analysis

1. Serial enzyme measurement
2. Isoenzymes determination
3. Measuring more than one enzyme

CINÉTICA

Serial measurement

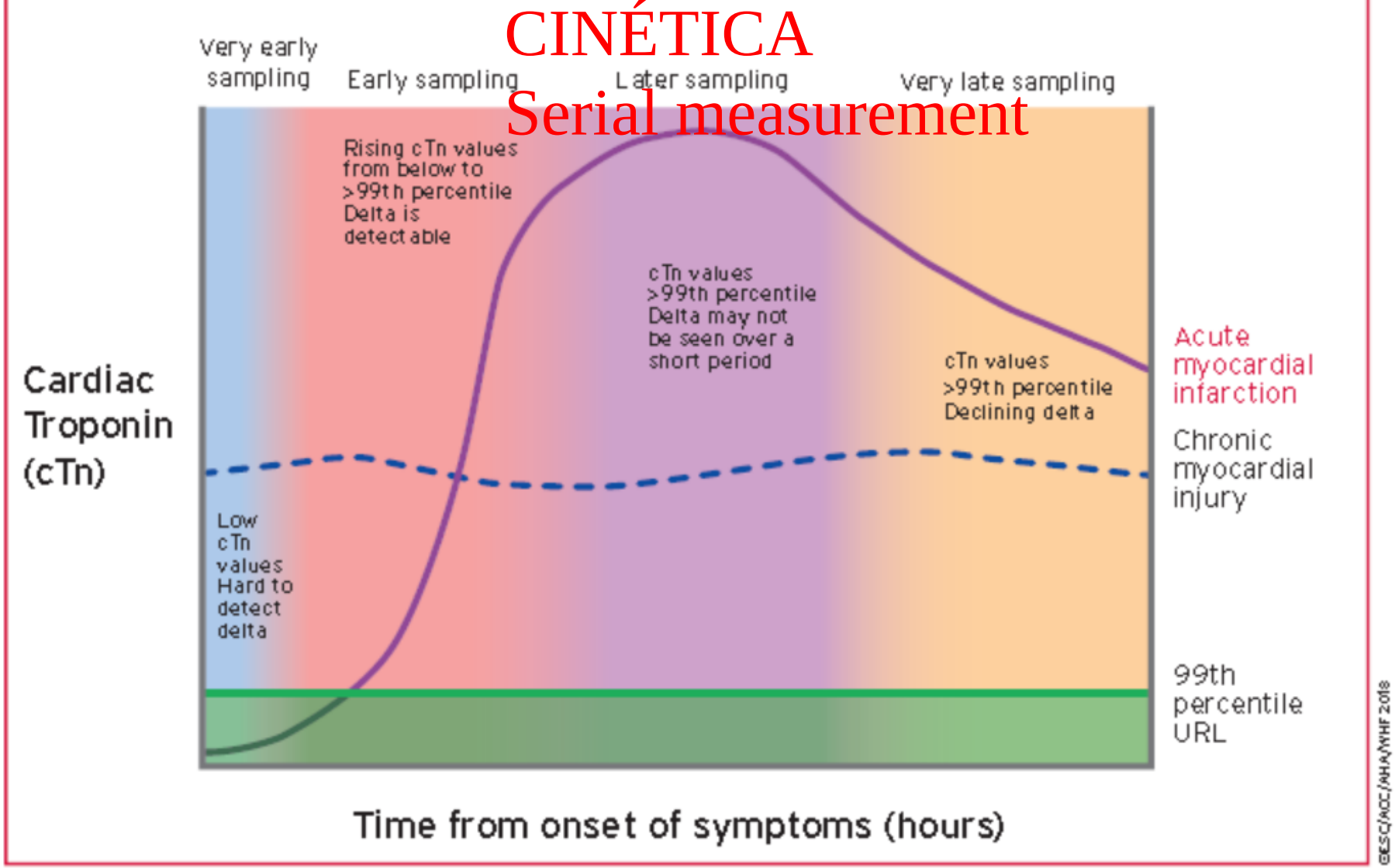
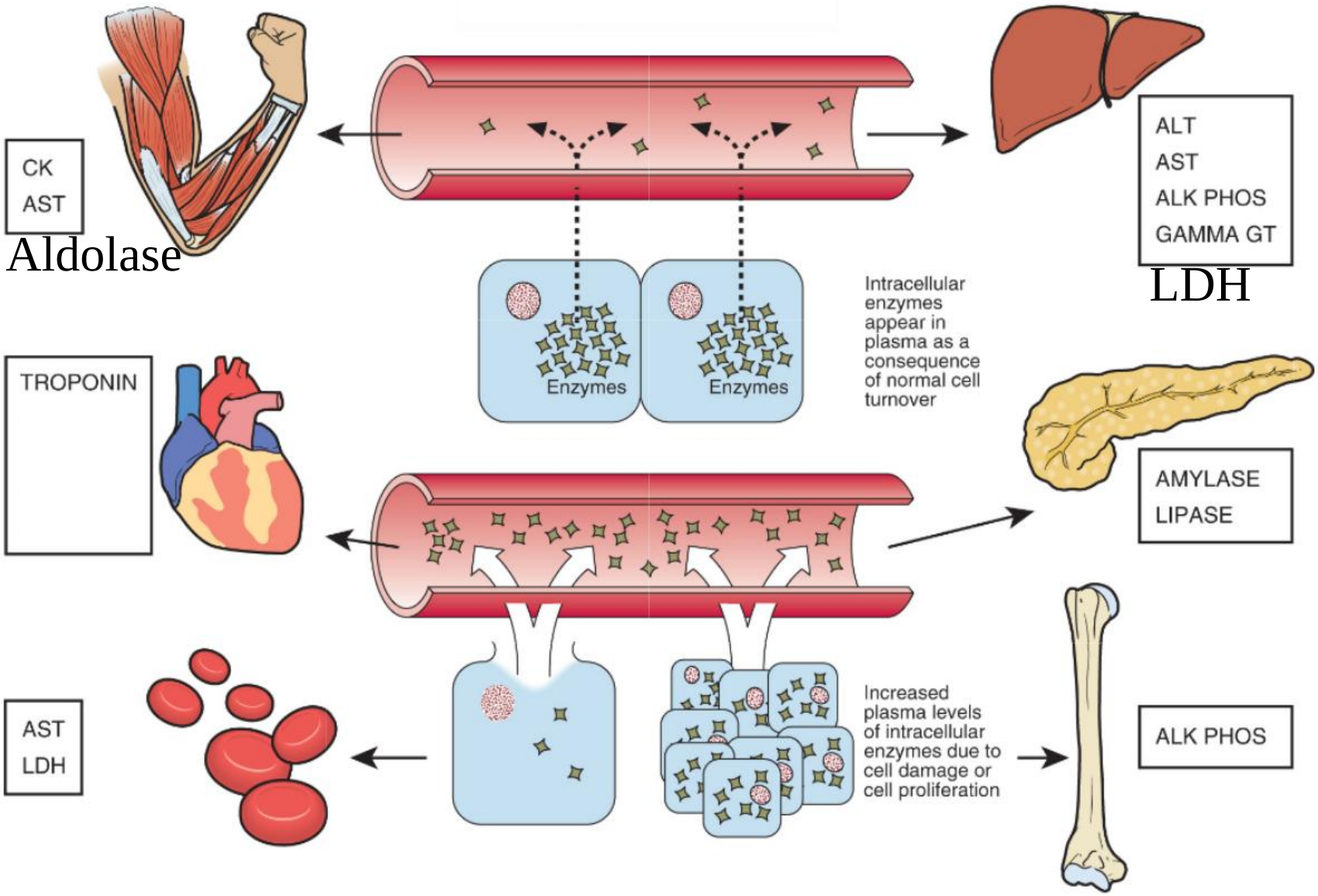


Figure 7 Illustration of early cardiac troponin kinetics in patients after acute myocardial injury including acute myocardial infarction. The timing of biomarker release into the circulation is dependent on blood flow and how soon after the onset of symptoms samples are obtained. Thus, the ability to consider small changes as diagnostic can be problematic. In addition, many comorbidities increase cTn values and, in particular, hs-cTn values, so that elevations can be present at baseline even in those with myocardial infarction who present early after the onset of symptoms. Changes in cTn values or deltas can be used to define acute compared with chronic events, and the ability to detect these is indicated in the figure. Increased cTn values can often be detected for days after an acute event. cTn – cardiac troponin; URL – upper reference limit.

3. Serum enzymes in disease

Enzymes and tissues



1. Liver enzymes

Aminotransferases

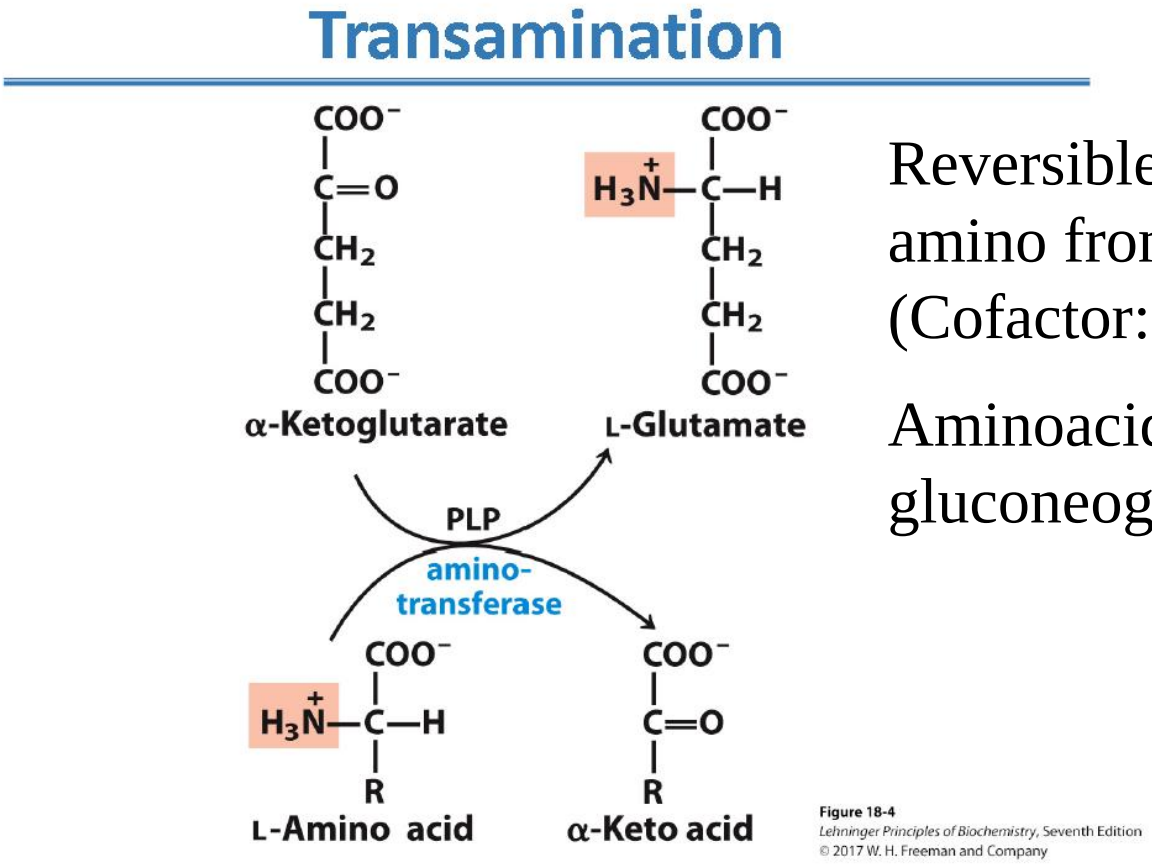


Figure 18-4
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Reversible transference of an amino from a α-aa to an α-ketoacid (Cofactor: piridoxal phosphate)

Aminoacid metabolism and gluconeogénesis

Aspartate aminotransferase and alanine aminotransferase

- AST aspartate + α-cetoglutarate ⟷ glutamate + oxalacetate
- ALT piruvate + glutamate ⟷ alanine + α-cetoglutarate

TABLA 17-1 Actividad enzimática relativa respecto al suero

	AST	ALT	CK
Hígado	7.100	2.850	1
Músculo esquelético	5.000	300	50.000
Corazón	7.800	450	10.000
Riñón	4.500	120	640
Eritrocitos	17	7	—

ALT: alanina-aminotransferasa; AST: aspartato-aminotransferasa; CK: creatina-cinasa.

Half life

AST: citosolic 17 h / mitochondrial 87 h

ALT: 45 h

Simultaneously quantified in clinical practice

Aminotransferases in liver diseases

- Hepatitis (virus, toxic)
  - AST y ALT > 10x reference value
  - If ALT is increased > 6 month: chronic
- Liver cirrosis
- Hepatocarcinoma
- Cholestasis (impaired flow of the bile From the liber to the duodenum. Also, elevation of alkaline phosphatase
- Drugs: statins, isotretinoin (Roacutan)

But do not forget that they are expressed in other tissues!!!

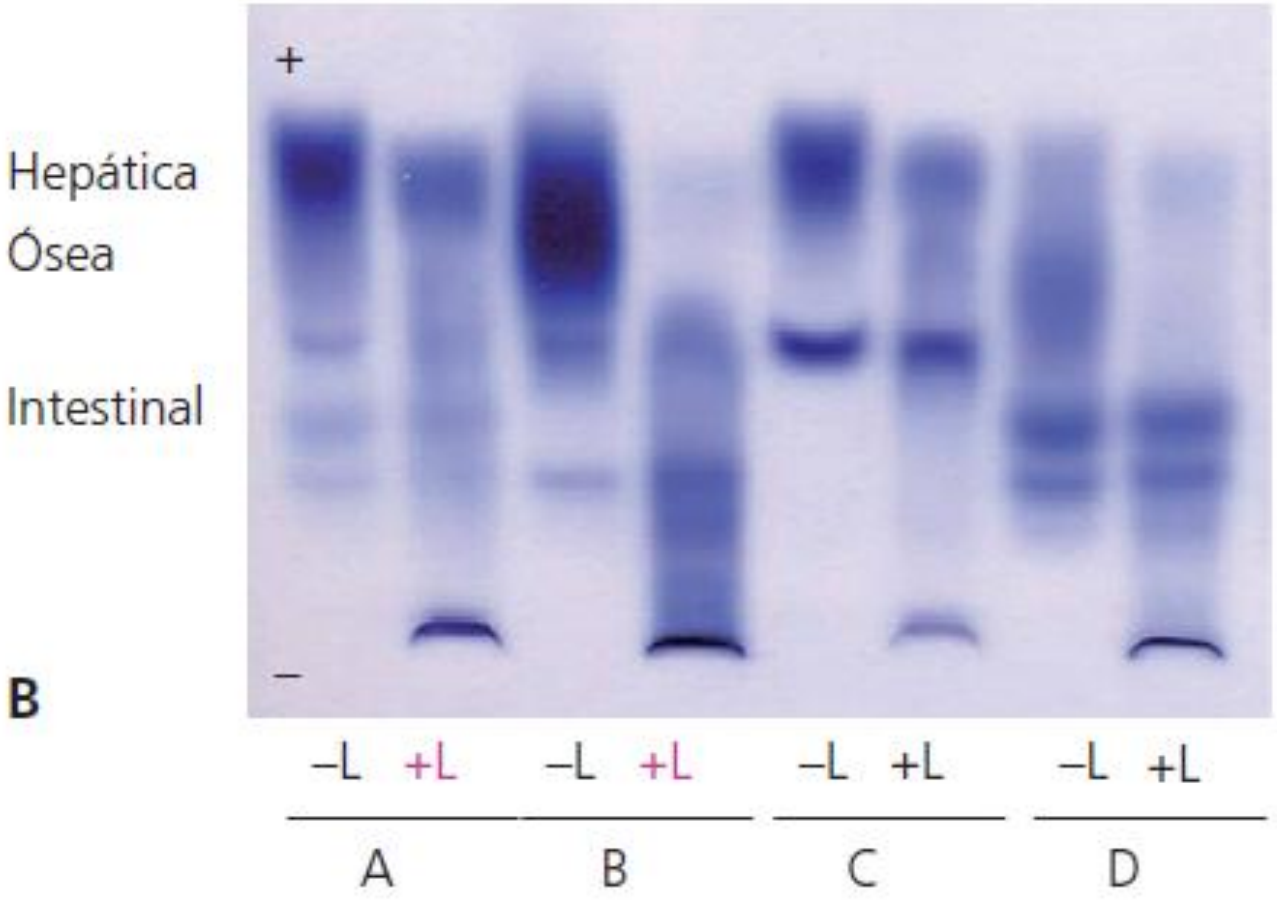
γ-Glutamiltransferase(γ-GT)

- Catalizes the transference of γ-glutamil
- Located in liver, kidney, pancreas and intestine
  - Bile duct cells
- Raise in serum
  - Cholestasis: more sensitive tan alkaline phosphatase
  - Alcohol intake
  - Drugs

Alkaline phosphatase

- Membrane metaloenzyme (Zn). Sample: no EDTA!!
- Role: dephosphorylates phosphate esters (Mg)
- optimally active at alkaline pH
- Location: sinusoids and bile canaliculi, bone, intestine, placenta,
- Requieres fasting
- There are isoenzymes -> Low specificity:
- Increase
  - Bone: Increase of osteoblastic activity: fractures, Vit D deficiency,
  - Liver: Bile tree, cholestasis

- Total ALP vs Isoenzymes separation



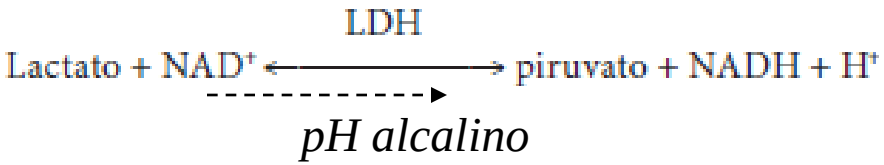
Lactate dehydrogenase (LDH)

- High molecular weight; tetramere: 2 types of subunits (H, M)

TABLA 17-3 Distribución de las isoenzimas de lactato-deshidrogenasa (LDH)

Isoenzima	Subunidades	Tejido en que predomina	Existencia en suero (%)	Ejemplo de incremento sérico
LD-1	H_4	Corazón y eritrocitos	17-31	Infarto de miocardio Hemólisis
LD-2	H_3M	Leucocitos	35-48	Leucemia
LD-3	H_2M_2	Pulmón	15-28	Embolismo pulmonar
LD-4	HM_3	Riñón y páncreas	5-9	Pancreatitis Necrosis renal
LD-5	M_4	Hígado y músculo esquelético	4-10	Enfermedad hepática

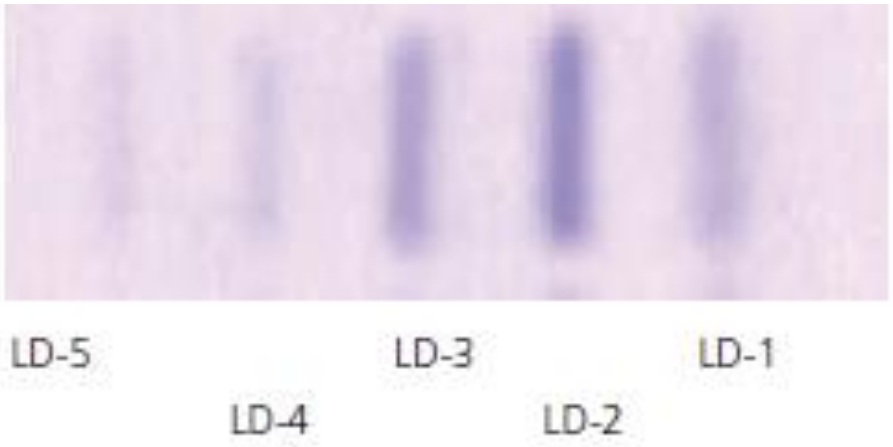
- Catalizes the reaction



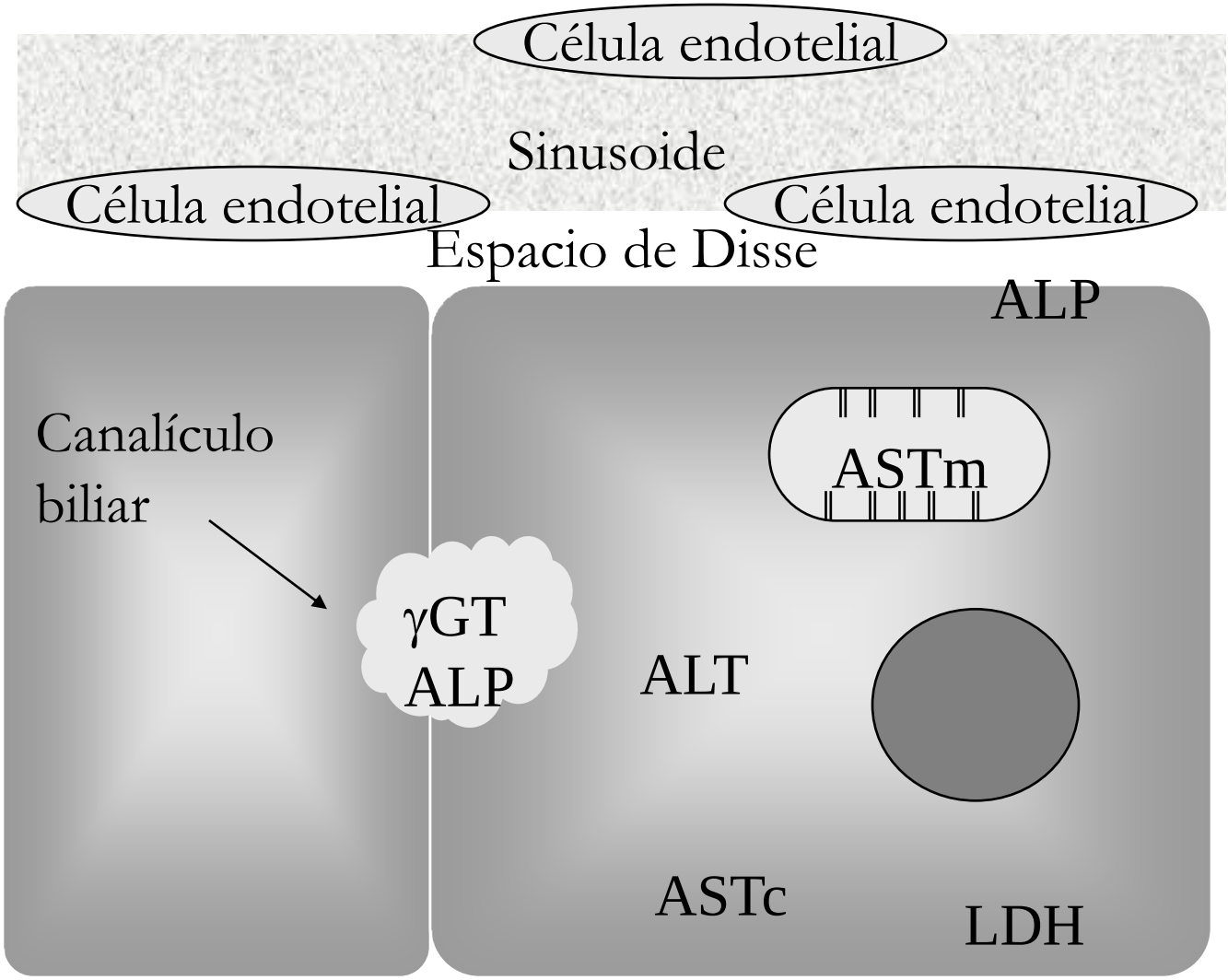
- Ubiquitous: Localized in all cells of the human body

Hemolysis!

- Synthesis in tumors: melanoma, leukemia... (↑LDH-5)
- Total LDH and Isoenzymes:

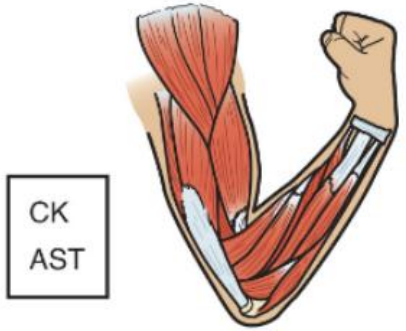


Enzymes as markers of hepatocytes integrity



2. Skeletal muscle enzymes

Creatin-kinase (CK)



- Cytosol enzyme

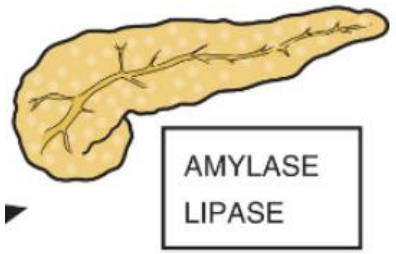
$$\text{Creatina} + \text{ATP} \underset{\text{CK/Mg2+ pH 6,7}}{\overset{\text{CK/Mg2+ pH 9}}{\longleftrightarrow}} \text{creatina-P} + \text{ADP.}$$

Do not memorize the reaction

- Dimer; 2 types of subunits (M and B)
  - CK-1 o CK-BB
  - CK-2 o CK-MB
  - CK-3 o CK-MM

- Tissue distribution
  - brain: 100% CK-BB
  - muscle: 97% CK-MM and 3% CK-MB
  - heart: 78% CK-MM and 22% CK-MB
- In blood, ussually 95% CK-MM
- Preanalytical variation: race, sex, age, exercise
- Increase
  1. Severe skeletal muscle lesions ↑↑CK (↑CK-MM)
     - Muscular distrophy, trauma, surgery, infections...
     - Not in nerological muscular diseases
  2. Cardiac damage
  3. Intense physical exercise

3. Pancreatic enzymes
a) Amylase



- Hydrolyses α-1,4 glycosidic bonds
- Metaloenzyme, low molecular weight, requires calcium
- No EDTA samples
- Amylase in urine
- Isoenzymes: Salivary (50%) and Pancreatic (50%). Not specific
- Hyperamylasemia in:
  - Acute pancreatitis:
    - Increase after 6 - 8 h of symptoms. (**Early** marker)
    - increase > 3 times reference values. No correlation with severity
    - Remains increased 3 - 5 days
  - Parotiditis, acute etanol intoxication...

b) Lipase

- Catalyzes the hydrolysis of glicerol and triglycerides esters
- Main source: páncreas. More **specific**
- Clinical use: Diagnosis of pancreatitis
  - Increase after 24 - 48 h of symptoms
  - increase > 5 times Reference Values
  - Enhanced 7 - 14 days

Acute pancreatitis

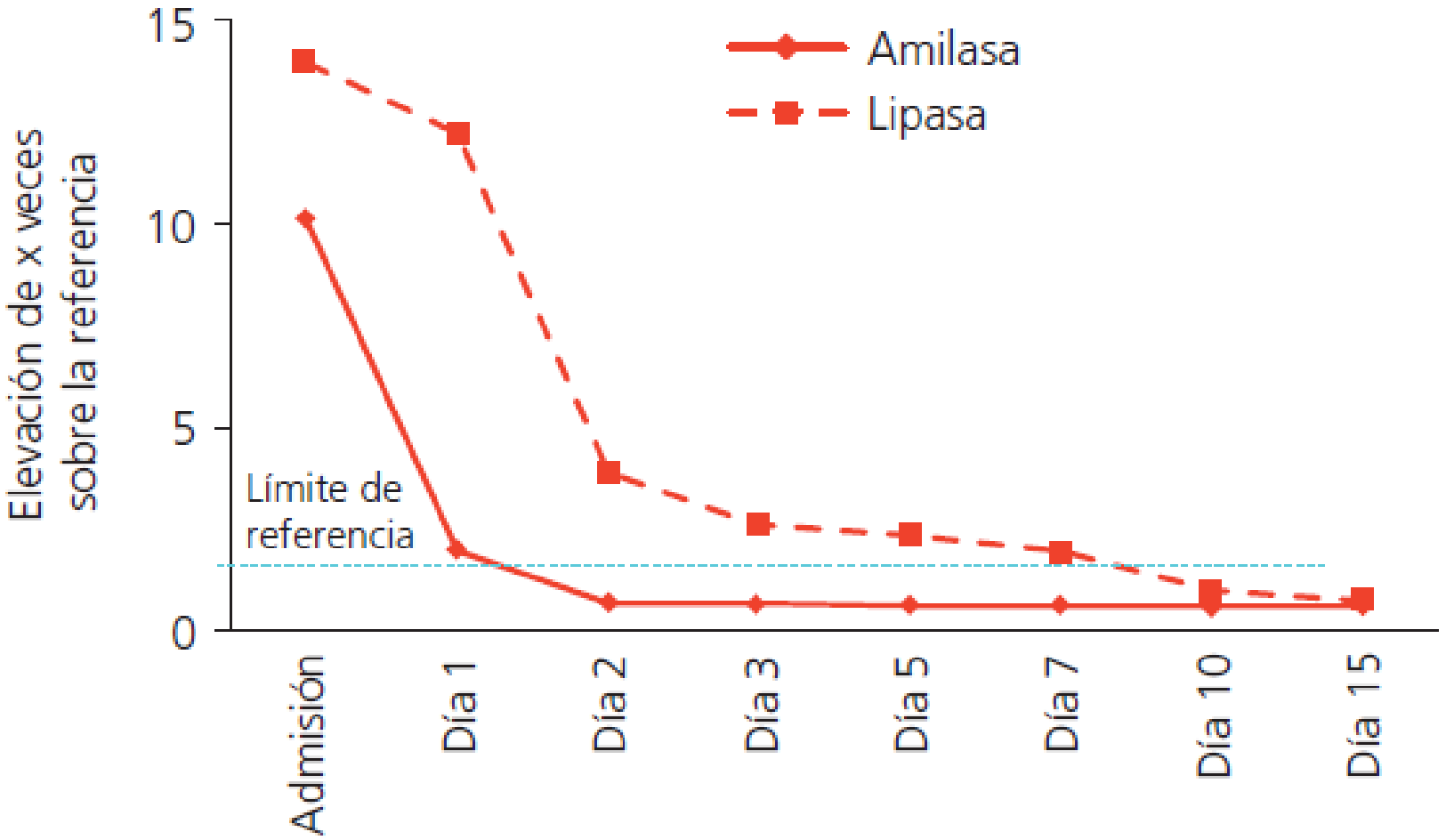
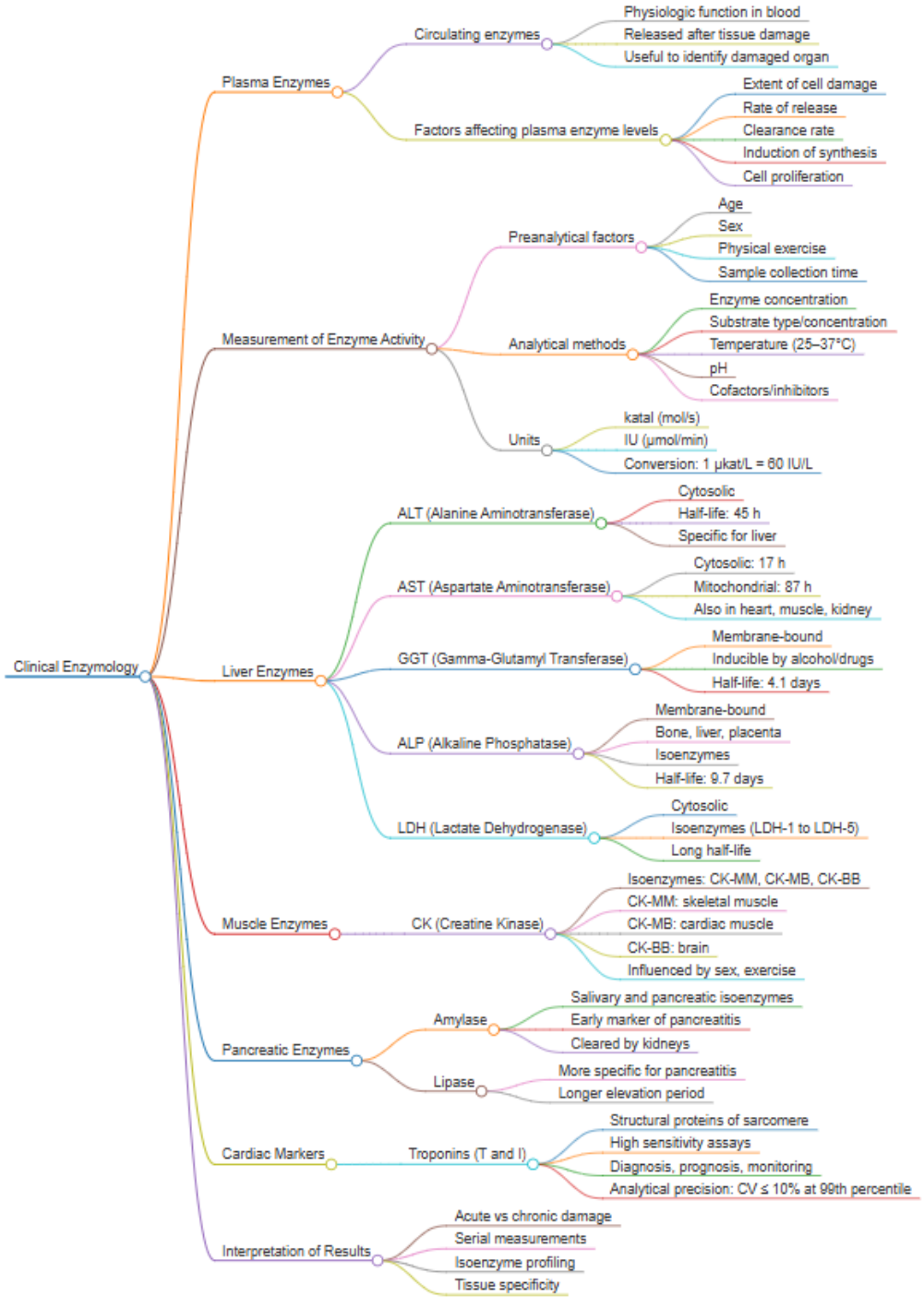


Figura 24-3 Ejemplo de la evolución de las actividades α-amilasa y lipasa durante un ingreso hospitalario por una pancreatitis aguda.

Enzyme	Primary Organ	Secondary Organs	Clinical Interpretation	Isoenzymes	Reference Values
ALT (Alanine Aminotransferase)	Liver	Kidney, heart, muscle	Elevated levels suggest hepatocellular injury. ALT is more liver-specific than AST. Used to monitor liver diseases such as hepatitis.	No	Slightly higher in men
AST (Aspartate Aminotransferase)	Liver	Heart, muscle, kidney, brain	Elevated in liver disease, myocardial infarction, and muscle injury. Less specific than ALT for liver damage.	No	Slightly higher in men
GGT (Gamma-Glutamyl Transferase)	Liver	Pancreas, kidney	Elevated in cholestasis and alcohol abuse. Sensitive marker for biliary tract disease.	No	Slightly higher in men
Alkaline Phosphatase (ALP)	Liver	Bone, placenta, intestine	Elevated in bone disorders (e.g., Paget's disease) and liver disease. Isoenzyme analysis helps differentiate source.	Yes (liver, bone, intestinal, placental)	Higher in children
LDH (Lactate Dehydrogenase)	Multiple tissues	Heart, liver, muscle, kidney, lungs	Elevated in tissue breakdown, hemolysis, myocardial infarction, and some cancers. Isoenzyme profiling can help localize source.	Yes (LDH-1 to LDH-5)	May vary with age
CK (Creatine Kinase)	Muscle	Heart, brain	Elevated in muscle injury, myocardial infarction, and rhabdomyolysis. CK-MB isoenzyme used for cardiac damage.	Yes (CK-MM, CK-MB, CK-BB)	Slightly higher in men
Amylase	Pancreas	Salivary glands	Elevated in acute pancreatitis and salivary gland disorders. Often measured with lipase.	Yes (pancreatic and salivary)	Similar in adults
Lipase	Pancreas	Stomach, intestine	More specific than amylase for pancreatitis. Elevated in pancreatic inflammation or obstruction.	No	Similar in adults
Troponin T/I	Heart	None	Highly specific marker for myocardial infarction. Rises within hours of cardiac injury and remains elevated for days.	Yes (Troponin T and I isoforms)	Varies by assay and age


Clinical Enzymology
Plasma Enzymes
Circulating enzymes
Physiologic function in blood
Released after tissue damage
Useful to identify damaged organ
Factors affecting plasma enzyme levels
Extent of cell damage
Rate of release
Clearance rate
Induction of synthesis
Cell proliferation
Measurement of Enzyme Activity
Preanalytical factors
Age
Sex
Physical exercise
Sample collection time
Analytical methods
Enzyme concentration
Substrate type/concentration
Temperature (25–37°C)
pH
Cofactors/inhibitors
Units
katal (mol/s)
IU (µmol/min)
Conversion: 1 µkat/L = 60 IU/L
Liver Enzymes
ALT (Alanine Aminotransferase)
Cytosolic
Half-life: 45 h
Specific for liver
AST (Aspartate Aminotransferase)
Cytosolic: 17 h
Mitochondrial: 87 h
Also in heart, muscle, kidney
GGT (Gamma-Glutamyl Transferase)
Membrane-bound
Inducible by alcohol/drugs
Half-life: 4.1 days
ALP (Alkaline Phosphatase)
Membrane-bound
Bone, liver, placenta
Isoenzymes
Half-life: 9.7 days
LDH (Lactate Dehydrogenase)
Cytosolic
Isoenzymes (LDH-1 to LDH-5)
Long half-life
Muscle Enzymes
CK (Creatine Kinase)
Isoenzymes: CK-MM, CK-MB, CK-BB
CK-MM: skeletal muscle
CK-MB: cardiac muscle
CK-BB: brain
Influenced by sex, exercise
Pancreatic Enzymes
Amylase
Salivary and pancreatic isoenzymes
Early marker of pancreatitis
Cleared by kidneys
Lipase
More specific for pancreatitis
Longer elevation period
Cardiac Markers
Troponins (T and I)
Structural proteins of sarcomere
High sensitivity assays
Diagnosis, prognosis, monitoring
Analytical precision: CV ≤ 10% at 99th percentile
Interpretation of Results
Acute vs chronic damage
Serial measurements
Isoenzyme profiling
Tissue specificity